

ST04

## Human Health, Diseases and Public Health Programmes

*Science and Technology is 18-23 questions a paper and health takes 4-6 of them. Two facts here are documented identical repeats - Plasmodium falciparum as the malignant form (2021 and 2023) and Trisomy 21 for Down syndrome (2018 and 2021). Vector-borne disease matching appears in 2013, 2021, 2023 and 2024.*

Disease, pathogen, vector - that triangle is the most reliably asked thing in the entire Science and Technology paper. RPSC has set a disease-to-organism or disease-to-vector match in every recent paper, and the second half of the chapter, the national health programmes, has become just as regular because RAS loves scheme names, launch years and target dates.

## Health, Disease and Health Programmes

### Vector-borne

- Malaria - Anopheles
- Dengue/chikungunya/Zika - Aedes
- Filariasis and JE - Culex
- Kala-azar - sandfly
- Plague - rat flea; scrub typhus - mite

### Deficiency

- A - night blindness
- B1 beriberi, B3 pellagra
- C scurvy, D rickets
- Iodine goitre, iron anaemia

### Programmes

- NHM = NRHM 2005 + NUHM 2013
- Ayushman Bharat 2018, PM-JAY Rs 5 lakh
- NTEP - TB by 2025
- ABDM - 14-digit ABHA
- NCVBDC - malaria by 2030

### Bacterial

- TB - Mycobacterium tuberculosis
- Leprosy - M. leprae
- Cholera - Vibrio cholerae
- Typhoid - Salmonella typhi, Widal test

### Genetic

- Down - trisomy 21, 47
- Turner 45 XO, Klinefelter 47 XXY
- Haemophilia, colour blindness - X-linked
- Sickle cell, thalassaemia - autosomal recessive

### Rajasthan angle

- Right to Health Act - first in India
- Chiranjeevi to Mukhyamantri Ayushman Arogya
- Nishulk Dawa and Janch Yojana
- POSHAN Abhiyaan launched from Jhunjhunu

### Viral

- Polio - Sabin OPV, Salk IPV
- Rabies - Pasteur, dog bite
- COVID-19 - SARS-CoV-2, 11 March 2020
- H1N1 swine, H5N1 bird
- Nipah - fruit bat

### Immunity

- Active vs passive
- Jenner 1796, smallpox out in 1980
- UIP 1985, 12 diseases
- Mission Indradhanush 25 Dec 2014
- Cold chain 2-8 degrees C

## How to read a disease question - the four classes of pathogen

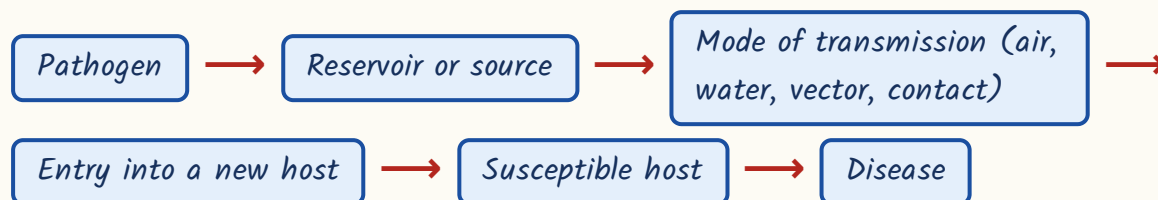
Before memorising anything, sort the diseases into four buckets by the kind of organism that causes them - virus, bacterium, protozoan, worm or fungus. Half the exam's negative-framed questions (which of the following is NOT a viral

disease') are settled by the bucket alone. Then ask the second question: does it need a vector to reach you, or does it come through air, water or contact?

### Pathogen class and the diseases that belong to it

| Class of pathogen | Diseases                                                                                                                    |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Virus             | COVID-19, polio, measles, mumps, chickenpox, rabies, influenza, dengue, chikungunya, hepatitis A-E, AIDS, Nipah, Zika, mpox |
| Bacterium         | Tuberculosis, leprosy, cholera, typhoid, plague, diphtheria, whooping cough, tetanus, pneumonia, scrub typhus               |
| Protozoan         | Malaria, kala-azar, amoebic dysentery, sleeping sickness                                                                    |
| Worm (helminth)   | Lymphatic filariasis ( <i>Wuchereria bancrofti</i> ), ascariasis, taeniasis                                                 |
| Fungus            | Ringworm, athlete's foot, candidiasis                                                                                       |

### The chain of infection - break any link and the disease stops



### The master table - disease, pathogen, vector

This is the centrepiece of the chapter. RPSC asks disease-to-organism or disease-to-vector matching every single year - 2013, 2021, 2023 and 2024 all carried one. Do not read these tables; recite them. Cover the second column and test yourself from the first, then reverse it.

### Vector-borne diseases - the highest-yield table in the chapter

| Disease                              | Pathogen                                  | Vector                                                                    |
|--------------------------------------|-------------------------------------------|---------------------------------------------------------------------------|
| Malaria                              | <i>Plasmodium</i> (protozoan)             | Female <i>Anopheles</i> mosquito                                          |
| Dengue                               | Dengue virus (a flavivirus)               | <i>Aedes aegypti</i> - a daytime biter                                    |
| Chikungunya                          | Chikungunya virus (alphavirus)            | <i>Aedes aegypti</i> and <i>Aedes albopictus</i>                          |
| Zika                                 | Zika virus                                | <i>Aedes aegypti</i>                                                      |
| Yellow fever                         | Yellow fever virus                        | <i>Aedes aegypti</i>                                                      |
| Japanese encephalitis                | JE virus                                  | <i>Culex tritaeniorhynchus</i> ; the PIG is the amplifying host           |
| Lymphatic filariasis (elephantiasis) | <i>Wuchereria bancrofti</i> (a nematode)  | <i>Culex quinquefasciatus</i>                                             |
| Kala-azar (visceral leishmaniasis)   | <i>Leishmania donovani</i> (protozoan)    | Female sandfly, <i>Phlebotomus argentipes</i>                             |
| Plague                               | <i>Yersinia pestis</i> (bacterium)        | Rat flea, <i>Xenopsylla cheopis</i>                                       |
| Scrub typhus                         | <i>Orientia tsutsugamushi</i> (bacterium) | Larval trombiculid mite (chigger); leaves an eschar at the bite           |
| African sleeping sickness            | <i>Trypanosoma</i> (protozoan)            | Tsetse fly                                                                |
| Kyasanur forest disease              | KFD virus                                 | Hard tick, <i>Haemaphysalis</i> - the 'monkey fever' of the Western Ghats |

### Diseases that need no vector - air, water, food and contact

| Disease                    | Pathogen                              | How it spreads                                       |
|----------------------------|---------------------------------------|------------------------------------------------------|
| Tuberculosis               | <i>Mycobacterium tuberculosis</i>     | Air-borne droplets                                   |
| Leprosy (Hansen's disease) | <i>Mycobacterium leprae</i>           | Prolonged close contact and droplets                 |
| Cholera                    | <i>Vibrio cholerae</i> (comma-shaped) | Contaminated water and food                          |
| Typhoid                    | <i>Salmonella typhi</i>               | Contaminated food and water; Widal test diagnoses it |

| Disease                    | Pathogen                                         | How it spreads                                      |
|----------------------------|--------------------------------------------------|-----------------------------------------------------|
| Diphtheria                 | <i>Corynebacterium diphtheriae</i>               | Droplets                                            |
| Whooping cough (pertussis) | <i>Bordetella pertussis</i>                      | Droplets                                            |
| Tetanus                    | <i>Clostridium tetani</i>                        | Spores from soil entering a wound - NOT water-borne |
| Pneumonia                  | <i>Streptococcus pneumoniae</i>                  | Droplets                                            |
| Amoebic dysentery          | <i>Entamoeba histolytica</i>                     | Contaminated food and water                         |
| Ascariasis                 | <i>Ascaris lumbricoides</i> (roundworm)          | Contaminated food and water                         |
| Ringworm                   | Fungi - <i>Trichophyton</i> , <i>Microsporum</i> | Contact, shared clothing, damp skin                 |
| Anthrax                    | <i>Bacillus anthracis</i>                        | Contact with infected animals or their products     |

**ASKED IN RAS**

Asked in RAS Pre 2024, twice over as matching sets - malaria with *Plasmodium* and *Anopheles*, kala-azar with *Leishmania donovani* and the sandfly, filariasis with *Wuchereria bancrofti* and *Culex*, typhoid with *Salmonella typhi*, dengue with *Aedes aegypti*. If you can reproduce those five rows you have the question.

**ASKED IN RAS**

Asked in RAS Pre 2024 as a NOT-matched question - 'Kala-azar : tsetse fly' is the wrong pair. Kala-azar is the SANDFLY; the tsetse fly carries African sleeping sickness.

**TRAP**

The three mosquitoes are where candidates lose the mark. Anopheles = malaria. Aedes = dengue, chikungunya, Zika, yellow fever (day biter, breeds in CLEAN stagnant water). Culex = filariasis and Japanese encephalitis. Also remember plague is bacterial though it has a vector, and scrub typhus is bacterial though it comes from a mite.

## Malaria in depth - the single most asked disease

Malaria is caused by a protozoan of the genus Plasmodium and is delivered by the bite of the female Anopheles mosquito - only the female, because only she takes a blood meal. The parasite multiplies first in the liver, then inside the red blood cells, and the cyclical fever with chills comes from the red cells bursting in batches.

### The Plasmodium species - the species question is the repeat

| Species               | Form of malaria                      | Point to remember                                           |
|-----------------------|--------------------------------------|-------------------------------------------------------------|
| Plasmodium falciparum | Malignant tertian / cerebral malaria | The most fatal form; accounts for almost all malaria deaths |
| Plasmodium vivax      | Benign tertian malaria               | The commonest species in India                              |
| Plasmodium malariae   | Quartan malaria                      | Fever every fourth day                                      |
| Plasmodium ovale      | Mild tertian malaria                 | Rare in India                                               |

- Ronald Ross demonstrated the malaria parasite inside an Anopheles mosquito at Secunderabad in 1897; Nobel Prize 1902.
- Anopheles is the definitive host of Plasmodium; the human being is the intermediate host.
- Diagnosis - blood smear microscopy or a rapid diagnostic test; treatment includes artemisinin-based combination therapy.

- Control - indoor residual spraying, long-lasting insecticidal nets, larvivorous fish such as *Gambusia*, and source reduction.
- India's target year for malaria elimination is 2030, under the National Center for Vector Borne Diseases Control.
- World Malaria Day is 25 April.

#### ASKED IN RAS

Asked in RAS Pre 2023, and again in 2021 - which species causes the most severe, malignant form of malaria? Answer: *Plasmodium falciparum*. The 2023 paper also ran it as a statement set where 'P. falciparum causes the benign tertian form' was the planted error, and as an assertion-reason: falciparum infects red cells of every age and the infected cells block cerebral capillaries.

#### YAAD RAKHO

Yaad rakho - 'Falciparum is Fatal, Vivax is Very common'. F for fatal and for cerebral; V for the vivax that dominates Indian case numbers.

### The other vector-borne diseases you must be able to name

After malaria, dengue is the one that reaches the newspapers every monsoon. It is a viral disease - a flavivirus - carried by *Aedes aegypti*, which bites in the daytime and breeds in clean stagnant water inside and around houses: coolers, tyres, flowerpots, overhead tanks. That is why source reduction, not fogging, is the real control measure.

- Dengue - high fever, severe joint and muscle pain ('breakbone fever'), rash, and a characteristic FALL in platelet count (thrombocytopenia).

- Severe forms are dengue haemorrhagic fever and dengue shock syndrome; there is no specific antiviral, treatment is supportive fluids.
- Chikungunya - the same Aedes vector, with crippling joint pain that can persist for months; rarely fatal.
- Zika - Aedes-borne, usually mild in adults, but infection in pregnancy can cause microcephaly in the newborn.
- Japanese encephalitis - Culex tritaeniorhynchus is the vector, the pig is the amplifying host, the ardeid water bird the natural reservoir; JE vaccine is given in endemic districts under UIP.
- Kala-azar - Leishmania donovani through the female sandfly Phlebotomus argentipes; prolonged fever with an enlarged spleen and darkening of the skin, which is what 'kala-azar' means.
- Lymphatic filariasis - the worm blocks the lymph vessels and the limb swells grossly, hence elephantiasis; mass drug administration is the control strategy.

### Malaria vs dengue - RPSC sets them against each other

| Malaria                                       | Dengue                                       |
|-----------------------------------------------|----------------------------------------------|
| Protozoan - Plasmodium                        | Virus - a flavivirus                         |
| Female Anopheles, bites at night              | Aedes aegypti, bites in the daytime          |
| Breeds in any stagnant water, including dirty | Breeds in CLEAN stored water                 |
| Cyclical fever with chills and rigors         | High fever with severe joint pain and rash   |
| Platelets usually normal                      | Platelet count falls - the signature finding |
| Curable with antimalarial drugs               | No specific antiviral; supportive care only  |

- The NVBDCP directorate is now the National Center for Vector Borne Diseases Control (NCVBDC).
- It covers SIX diseases - malaria, dengue, chikungunya, Japanese encephalitis, kala-azar and lymphatic filariasis.
- Only THREE of them are targeted for elimination - malaria, kala-azar and lymphatic filariasis.

- Dengue, chikungunya and JE are outbreak-prone and are managed by control, not elimination - a favourite statement trap.
- Tuberculosis is not part of this programme at all; it has its own, the NTEP.

## Tuberculosis, leprosy and the water-borne classics

Tuberculosis is India's biggest single infectious disease burden and its exam questions are settled by four names: Robert Koch, who identified *Mycobacterium tuberculosis* in 1882; BCG, the live attenuated vaccine given at birth; DOTS, the directly observed treatment; and Ni-kshay, the digital notification portal. World TB Day is 24 March, the day Koch announced the discovery.

- TB is air-borne, spreads by droplets, and most often attacks the lungs (pulmonary TB), though it can hit bone, spine, brain or intestine.
- Diagnosis - sputum microscopy, CBNAAT (a rapid molecular test) and chest X-ray.
- BCG (*Bacillus Calmette-Guerin*) is LIVE ATTENUATED, not killed - it protects mainly against severe childhood TB, miliary TB and TB meningitis.
- Multi-drug resistant TB is resistance to at least isoniazid and rifampicin.
- Leprosy - *Mycobacterium leprae*, discovered by the Norwegian physician G.H.A. Hansen in 1873; treated with multi-drug therapy, free of cost.
- Cholera - *Vibrio cholerae*, profuse painless watery diarrhoea; oral rehydration solution is the life-saver.
- Typhoid - *Salmonella typhi*, step-ladder fever, diagnosed by the Widal test; typhoid vaccine exists but is not in the routine UIP schedule.

## Hepatitis A to E - the classic sorting question

| Type        | How it spreads                           | Vaccine           |
|-------------|------------------------------------------|-------------------|
| Hepatitis A | Faeco-oral - contaminated food and water | Vaccine available |

| Type        | How it spreads                      | Vaccine                                                  |
|-------------|-------------------------------------|----------------------------------------------------------|
| Hepatitis B | Blood, body fluids, mother to child | Vaccine available and part of the UIP, with a birth dose |
| Hepatitis C | Blood and blood products            | No vaccine; curable with modern antivirals               |
| Hepatitis D | Only along with hepatitis B         | Prevented by the hepatitis B vaccine                     |
| Hepatitis E | Faeco-oral, mainly water            | Especially dangerous in pregnancy                        |

**TRAP**

Split them two and three. A and E travel by WATER (the two vowels at the ends). B, C and D travel by BLOOD. If a question says hepatitis B spreads through contaminated water, it is wrong; if it says hepatitis A has a vaccine in the UIP, that is also wrong - it is hepatitis B that is in the UIP.

## The viral diseases - polio, rabies, COVID-19 and the newer ones

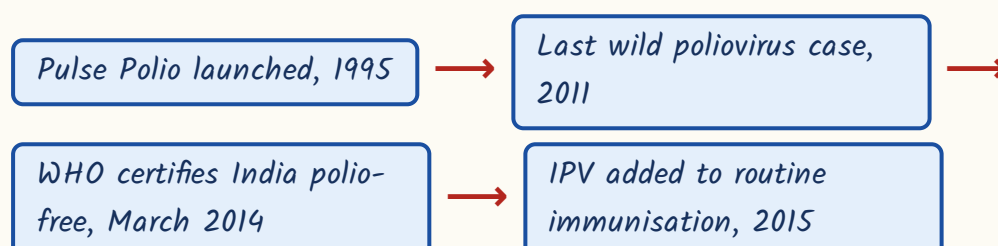
Viruses give the exam its dates. Polio gives you 1955 and 2014, smallpox 1796 and 1980, COVID-19 gives 11 March 2020. Learn the vaccine names attached to each, because the vaccine-to-disease pairing is asked as often as the disease itself.

- Polio - poliovirus. OPV is the LIVE attenuated oral vaccine of Albert Sabin; IPV is the KILLED injectable vaccine of Jonas Salk.
- Pulse Polio Immunisation began in India in 1995; injectable polio vaccine was added to routine immunisation in 2015.
- India was certified polio-free by the WHO in March 2014.
- Smallpox - Edward Jenner used cowpox material in 1796; WHO declared the disease eradicated in 1980, India's last case being in 1975. It is the only

human disease so far eradicated.

- Rabies - a lyssavirus, spread by the bite or lick of an infected animal, most often a dog. Louis Pasteur made the first vaccine, 1885. Once symptoms appear it is virtually 100 per cent fatal, so post-exposure prophylaxis must not be delayed.
- Measles - a virus; measles-rubella (MR) vaccine is given under the UIP at 9-12 months and again at 16-24 months.
- COVID-19 - caused by SARS-CoV-2; the WHO declared it a pandemic on 11 March 2020.
- India's principal vaccines were Covaxin (Bharat Biotech with ICMR-NIV, inactivated) and Covishield (Serum Institute, the Oxford-AstraZeneca viral-vector vaccine).
- Influenza - swine flu is influenza A H1N1 (the 2009 pandemic strain); bird flu is avian influenza H5N1.
- Nipah - the natural reservoir is the fruit bat of the genus *Pteropus*, with pigs as intermediate hosts; India has seen outbreaks in West Bengal and Kerala.
- Mpox - a viral zoonosis of the orthopoxvirus group; the WHO declared it a public health emergency of international concern in August 2024.
- AIDS - HIV, a retrovirus that destroys the CD4 helper T lymphocytes; screened by ELISA and confirmed by Western blot.
- NACO was set up in 1992 and the HIV and AIDS (Prevention and Control) Act was passed in 2017. World AIDS Day is 1 December.

### India's polio timeline - a ready-made ordering question



## Deficiency diseases, lifestyle diseases and genetic disorders

Not every disease comes from an organism. Some come from what is missing in the plate, some from how we live, and some from the chromosomes we are born with. The deficiency table is pure recall and RPSC has asked it repeatedly as a match-the-following; the genetic table is where Down syndrome sits, and Down syndrome has been asked in identical form in 2018 and 2021.

### Deficiency diseases - nutrient to disease

| Nutrient                     | Deficiency disease                                | Note                                                                        |
|------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------|
| Vitamin A<br>(retinol)       | Night blindness,<br>xerophthalmia                 | Fat-soluble; a leading preventable cause of childhood blindness             |
| Vitamin B1<br>(thiamine)     | Beriberi                                          | Polished rice diets are the classic cause                                   |
| Vitamin B2<br>(riboflavin)   | Cheilosis, angular<br>stomatitis                  | Cracks at the corners of the mouth                                          |
| Vitamin B3<br>(niacin)       | Pellagra                                          | The three D's - dermatitis, diarrhoea, dementia                             |
| Vitamin B12<br>(cobalamin)   | Pernicious anaemia                                | The only vitamin containing cobalt                                          |
| Vitamin C<br>(ascorbic acid) | Scurvy                                            | Bleeding gums, delayed wound healing - collagen fails                       |
| Vitamin D<br>(calciferol)    | Rickets in children,<br>osteomalacia in adults    | The sunshine vitamin                                                        |
| Vitamin K                    | Delayed blood clotting                            | Needed for prothrombin                                                      |
| Iodine                       | Goitre, cretinism in infants                      | Universal salt iodisation is the answer                                     |
| Iron                         | Anaemia                                           | Addressed by Anaemia Mukht Bharat                                           |
| Calcium                      | Rickets, osteoporosis, tetany                     | Works with vitamin D                                                        |
| Fluoride                     | Deficiency - dental caries;<br>EXCESS - fluorosis | Fluorosis is endemic in parts of Rajasthan; NPPCF is the national programme |

| Nutrient           | Deficiency disease       | Note                                         |
|--------------------|--------------------------|----------------------------------------------|
| Protein and energy | Kwashiorkor and marasmus | The two faces of protein-energy malnutrition |

- *Non-communicable or lifestyle diseases - diabetes, hypertension, cardiovascular disease, cancer, chronic lung disease and obesity.*
- *They now cause the majority of deaths in India; the national programme for them is NPCDCS, run through district NCD clinics.*
- *Type 1 diabetes is insulin deficiency, type 2 is insulin resistance; obesity is measured by Body Mass Index.*
- *Hypertension is called the 'silent killer' because it usually has no symptoms until damage is done.*
- *World Health Day 7 April, World Cancer Day 4 February, World Blood Donor Day 14 June, Indian Organ Donation Day 3 August.*

### *Genetic disorders - the basis is the answer*

| Disorder             | Genetic basis                                    | Key point                                                                             |
|----------------------|--------------------------------------------------|---------------------------------------------------------------------------------------|
| Down syndrome        | Trisomy of chromosome 21 - 47 chromosomes in all | Intellectual disability, characteristic facial features; risk rises with maternal age |
| Turner syndrome      | 45, XO                                           | A female with a single X - short stature, sterile                                     |
| Klinefelter syndrome | 47, XXY                                          | A male with an extra X - sterile                                                      |
| Haemophilia          | X-linked recessive                               | Blood fails to clot; the 'royal disease', far commoner in males                       |
| Colour blindness     | X-linked recessive                               | Usually red-green; tested by Ishihara charts                                          |
| Sickle cell anaemia  | Autosomal recessive (haemoglobin S)              | National Sickle Cell Anaemia Elimination Mission, 2023, targets 2047                  |
| Thalassaemia         | Autosomal recessive                              | Defective haemoglobin chains; needs repeated transfusion                              |

| Disorder        | Genetic basis       | Key point                                       |
|-----------------|---------------------|-------------------------------------------------|
| Phenylketonuria | Autosomal recessive | The enzyme phenylalanine hydroxylase is missing |

**ASKED IN RAS**

Asked in RAS Pre 2018 and again, identically, in RAS Pre 2021 - Down syndrome is caused by trisomy of chromosome 21. Learn the number 21 and the resulting total of 47 chromosomes; both have been used as the option.

**TRAP**

Thalassaemia and sickle cell anaemia are AUTOSOMAL recessive - they are not sex-linked. Only haemophilia and colour blindness are the X-linked pair. RPSC plants 'thalassaemia is a sex-linked disorder' as the false statement.

## Immunity, vaccines, the UIP and antimicrobial resistance

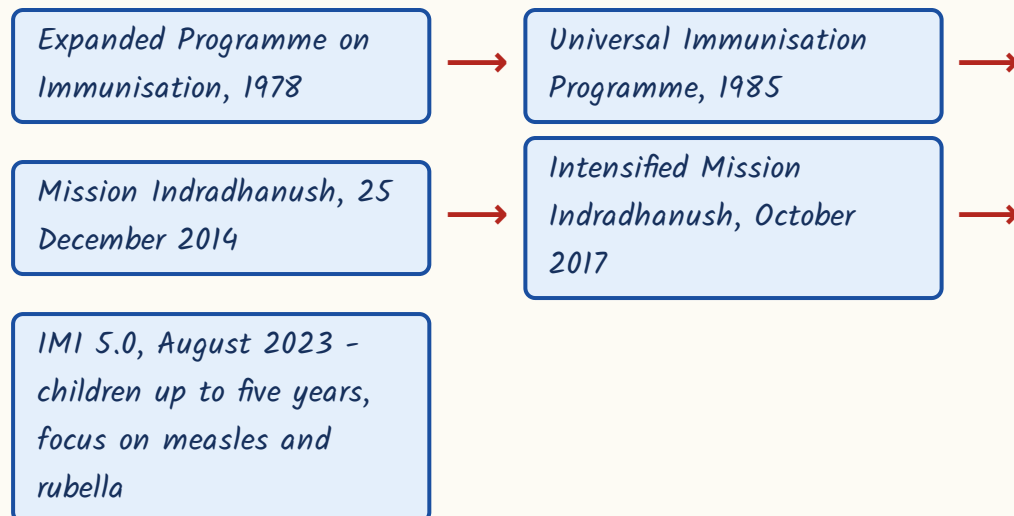
Immunity is either innate - the skin, the stomach acid, the general defence you are born with - or acquired, which is built after exposure. Acquired immunity then splits again: active, where your own body makes the antibodies, and passive, where ready-made antibodies are handed to you. Almost every immunity question in RAS is testing that one split.

### Active immunity vs passive immunity

| <b>Active immunity</b>                        | <b>Passive immunity</b>                        |
|-----------------------------------------------|------------------------------------------------|
| The body makes its own antibodies             | Ready-made antibodies come from outside        |
| Follows an infection or a vaccination         | Colostrum, antiserum, anti-D, anti-snake venom |
| Slow to develop                               | Acts immediately                               |
| Long-lasting, because memory cells are formed | Short-lived, no memory                         |
| Example - BCG, OPV, hepatitis B vaccine       | Example - mother's milk, anti-rabies serum     |

- Vaccine types - live attenuated (BCG, OPV, measles), killed or inactivated (IPV, Covaxin), toxoid (tetanus, diphtheria), subunit and recombinant (hepatitis B), and the newer mRNA and viral-vector vaccines.
- BCG is live attenuated; saying 'BCG is a killed vaccine' is a standard planted error.
- The Universal Immunisation Programme was launched in 1985, growing out of the Expanded Programme on Immunisation of 1978.
- The UIP now protects against 12 vaccine-preventable diseases - diphtheria, pertussis, tetanus, polio, measles, rubella, childhood tuberculosis, hepatitis B, Haemophilus influenzae type b infections, rotavirus diarrhoea, pneumococcal pneumonia, and Japanese encephalitis in endemic districts.
- Vaccines are kept in the cold chain at 2 to 8 degrees Celsius; breaking the cold chain destroys the vaccine.
- At birth a child gets BCG, the zero dose of OPV and the birth dose of hepatitis B vaccine.

### How India's immunisation drive grew



- Mission Indradhanush aims at full immunisation coverage of at least 90 per cent among children and pregnant women; the name comes from the rainbow's seven colours, the original seven vaccine-preventable diseases.
- Antibiotics work on bacteria only - they do nothing against a virus, so they are useless for a common cold.
- Alexander Fleming discovered penicillin, the first antibiotic, in 1928.
- Antimicrobial resistance grows from over-prescription, self-medication, unfinished courses and heavy antibiotic use in poultry and livestock.
- India adopted a National Action Plan on Antimicrobial Resistance in 2017; the Red Line campaign marks prescription-only antibiotics with a red vertical line on the pack.

### National health programmes, the indicators, and Rajasthan's Right to Health

This half of the chapter is pure scheme recall - name, launch year, and the one number attached to it. RAS asks it as matching or as a chronology, and it asks

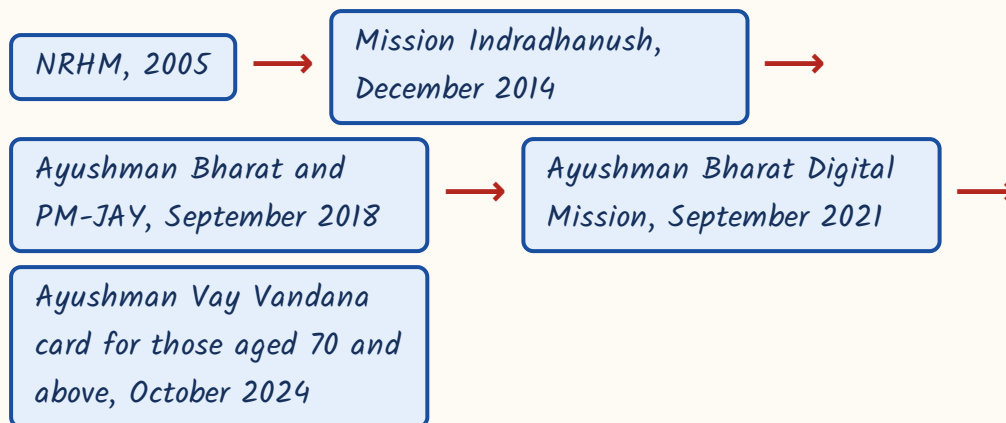
it in current affairs too. Learn the table with the years, then learn the four or five numbers that follow it.

### National health programmes - name, year, and what it actually does

| Programme                         | Launched                                         | What it does                                                                                 |
|-----------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------|
| National Rural Health Mission     | 2005                                             | Rural health system; created the ASHA, the community link worker                             |
| Janani Suraksha Yojana            | 2005                                             | Cash assistance for institutional delivery                                                   |
| Janani Shishu Suraksha Karyakram  | 2011                                             | Free delivery, drugs, diagnostics, diet and transport                                        |
| National Urban Health Mission     | 2013                                             | Urban health; NRHM + NUHM together make the National Health Mission                          |
| Mission Indradhanush              | 25 December 2014                                 | Full immunisation coverage of at least 90 per cent                                           |
| LaQshya                           | 2017                                             | Quality of care in the labour room and maternity operation theatre                           |
| Ayushman Bharat                   | 2018                                             | Two arms - Health and Wellness Centres plus PM-JAY                                           |
| PM-JAY                            | 23 September 2018, from Ranchi                   | Rs 5 lakh per family per year, secondary and tertiary care, no premium                       |
| Ni-kshay Poshan Yojana            | April 2018                                       | Nutrition support to TB patients by DBT, raised to Rs 1,000 a month in November 2024         |
| Anaemia Mukta Bharat              | 2018                                             | The 6x6x6 strategy - six beneficiary groups, six interventions, six institutional mechanisms |
| POSHAN Abhiyaan                   | 8 March 2018, launched from Jhunjhunu, Rajasthan | Targets stunting, anaemia and low birth weight                                               |
| National TB Elimination Programme | Renamed from RNTCP in 2020                       | TB elimination by 2025, five years ahead of the SDG target of 2030                           |

| Programme                       | Launched                                      | What it does                                                                               |
|---------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------|
| Ayushman Bharat Digital Mission | 27 September 2021<br>(piloted 15 August 2020) | The 14-digit ABHA number, plus the Healthcare Professionals and Health Facility Registries |
| NCVBDC (from NVBDCP)            | Renamed directorate                           | Six vector-borne diseases; malaria elimination target 2030                                 |

### Chronology of the big health initiatives - asked as an ordering question



- Health and Wellness Centres were renamed Ayushman Arogya Mandir in November 2023, with the tagline 'Arogyam Parmam Dhanam'.
- PM-JAY is fully government-funded - beneficiaries pay no premium; families were identified from the deprivation criteria of SECC 2011.
- The Ayushman Vay Vandana card, announced in October 2024, extends cover to all citizens aged 70 and above regardless of income.
- The Pradhan Mantri Bhartiya Janaushadhi Pariyojana (generic medicine stores) is a SEPARATE scheme - it is not a component of Ayushman Bharat.
- NOTTO, the National Organ and Tissue Transplant Organisation, sits at New Delhi under the DGHS, with ROTTOs and SOTTOs below it, under the Transplantation of Human Organs Act, 1994, amended in 2011.
- AYUSH covers Ayurveda, Yoga and Naturopathy, Unani, Siddha, Sowa-Rigpa and Homoeopathy; the Ministry was created in November 2014 and the WHO Global Centre for Traditional Medicine is at Jamnagar, Gujarat.

- Other standing programmes - National Programme for Control of Blindness and Visual Impairment, National Mental Health Programme (1982, with the Mental Healthcare Act 2017 and Tele-MANAS), and NPPCF for fluorosis.
- Health indicators, from the SRS Report 2021 released in May 2025 - maternal mortality ratio 93 per lakh live births (2019-21), infant mortality rate 27, under-five mortality rate 31 and neonatal mortality rate 19 per 1,000 live births.
- MMR is counted per lakh live births; IMR, USMR and NMR are per thousand. Mixing up the denominator is the standard error.
- Life expectancy, sex ratio at birth and total fertility rate are the other indicators RAS picks from.

#### CURRENT LINK

Current link, as of September 2026 - Rajasthan is the FIRST state in India to give its residents a legal right to health. The Rajasthan Right to Health Act was passed by the Assembly on 21 March 2023 and received the Governor's assent on 12 April 2023 (Act No. 7 of 2023). Its core promise is emergency treatment and care at any health-care establishment, public or private, without waiting for prepayment or police clearance. It sets up a State Health Authority and District Health Authorities, with a grievance to be answered within 24 hours and an action-taken report within 30 days. Note also that the state's Chiranjeevi Swasthya Bima Yojana (2021) has been renamed the Mukhyamantri Ayushman Arogya Yojana, with cover raised to Rs 25 lakh per family per year - check the latest state budget for the current figure before the exam.

- Rajasthan's older health flagships still get asked - Mukhyamantri Nishulk Dawa Yojana (free medicines, 2011) and Mukhyamantri Nishulk Janch Yojana (free diagnostics, 2013).
- Nirogi Rajasthan is the state's preventive-health campaign.

- POSHAN Abhiyaan was launched from Jhunjhunu in Rajasthan on 8 March 2018 - a Rajasthan link worth carrying into the current-affairs paper too.

### YAAD RAKHO

Yaad rakho the year ladder for the schemes - 2005 NRHM and JSY, 2011 JSSK, 2013 NUHM, 2014 Mission Indradhanush, 2017 LaQshya, 2018 Ayushman Bharat plus Ni-kshay Poshan plus Anaemia Mukht Bharat plus POSHAN, 2020 NTEP, 2021 ABDM. Almost every year has exactly one flagship - and 2018 has four.

### REVISION RECAP

1. Malaria - Plasmodium, female Anopheles; falciparum = malignant and cerebral, vivax = commonest in India; Ronald Ross 1897, Nobel 1902.
2. Aedes aegypti (day biter, clean water) carries dengue, chikungunya, Zika and yellow fever; dengue drops the platelet count.
3. Culex carries filariasis (Wuchereria bancrofti) and Japanese encephalitis, whose amplifying host is the pig.
4. Kala-azar = Leishmania donovani + sandfly. Plague = Yersinia pestis + rat flea. Scrub typhus = Orientia + chigger mite. Sleeping sickness = Trypanosoma + tsetse fly.
5. TB - Mycobacterium tuberculosis, Koch 1882, air-borne, BCG (live attenuated), CBNAAT, DOTS, Ni-kshay; World TB Day 24 March.
6. Leprosy = M. leprae (Hansen); cholera = Vibrio cholerae; typhoid = Salmonella typhi, diagnosed by the Widal test.
7. Hepatitis A and E through water; B, C and D through blood; only hepatitis B is in the UIP.
8. Polio - OPV live (Sabin), IPV killed (Salk); Pulse Polio 1995, India polio-free March 2014. Rabies - lyssavirus from a dog bite, Pasteur's vaccine, near-certain death once symptoms start. Smallpox - Jenner 1796, eradicated 1980.
9. COVID-19 = SARS-CoV-2, pandemic declared 11 March 2020; swine flu H1N1, bird flu H5N1; Nipah reservoir is the fruit bat; mpox is an orthopoxvirus.
10. HIV is a retrovirus destroying CD4 T cells - ELISA screens, Western blot confirms; NACO 1992, HIV/AIDS Act 2017.
11. Deficiencies - A night blindness, B1 beriberi, B3 pellagra, B12 pernicious anaemia, C scurvy, D rickets, K clotting, iodine goitre, iron anaemia; excess fluoride gives fluorosis, a Rajasthan problem.

12. Down = trisomy 21 (47 chromosomes); Turner 45 XO; Klinefelter 47 XXY; haemophilia and colour blindness X-linked recessive; sickle cell and thalassaemia autosomal recessive.
13. Active immunity is made by the body and lasts; passive immunity is borrowed and fades. BCG is live, not killed.
14. UIP 1985, from the EPI of 1978, now covers 12 diseases; cold chain 2-8 degrees C; Mission Indradhanush 25 December 2014, IMI 5.0 August 2023.
15. Antibiotics do not touch viruses; Fleming found penicillin in 1928; India's National Action Plan on AMR dates from 2017, plus the Red Line campaign.
16. NHM = NRHM (2005) + NUHM (2013); the ASHA is the NRHM's link worker.
17. Ayushman Bharat 2018 = Ayushman Arogya Mandir (the renamed HWCs, November 2023) + PM-JAY (Rs 5 lakh, launched 23 September 2018 from Ranchi, no premium).
18. NTEP targets TB elimination by 2025; NCVBDC covers six diseases and targets malaria elimination by 2030; ABDM gives the 14-digit ABHA number.
19. SRS 2021 - MMR 93 per lakh, IMR 27, USMR 31, NMR 19 per thousand.
20. Rajasthan passed India's FIRST Right to Health Act - Assembly 21 March 2023, assent 12 April 2023 - guaranteeing emergency care without prepayment; Chiranjeevi is now Mukhyamantri Ayushman Arogya Yojana.